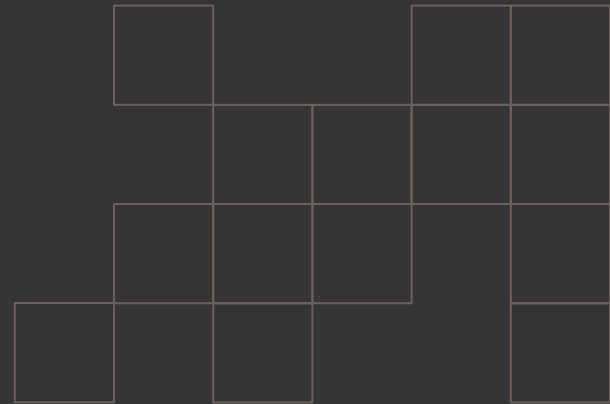


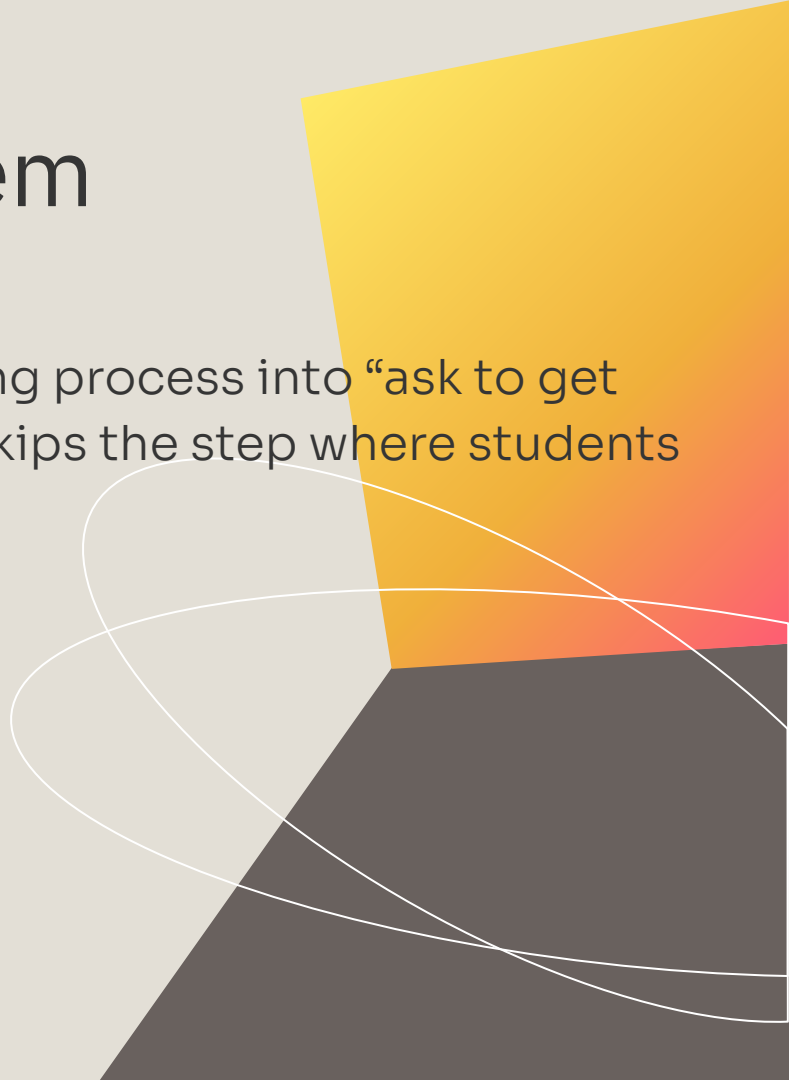
Team 8 GR5 Low-Fi Prototyping

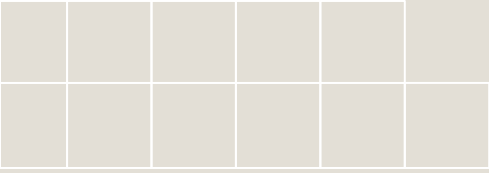




The Problem

Current AI tutoring often collapses the learning process into “ask to get answer,” which reduces student agency and skips the step where students externalize and debug their own reasoning.

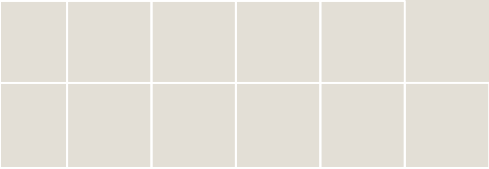




Value Proposition

A Stimulating, Personalized Guide for the Modern Age





The Solution

We solve this with a neutral, adaptive AI textbook that gives small, digestible hints and then asks targeted follow-up questions so the student actively builds the explanation themselves.

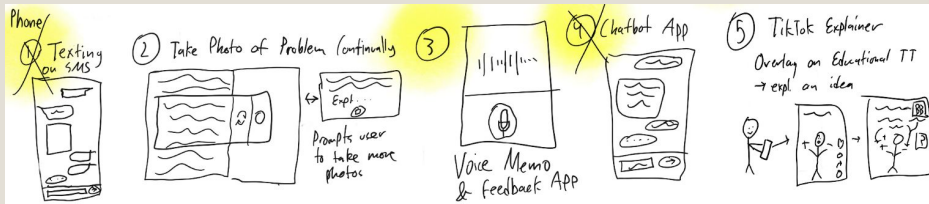


Contents

1.
Brainstorming, Sketches, Key Frames
2.
Task Flows
3.
Testing Methodology
4.
Testing Results
5.
Discussion
6.
Appendix

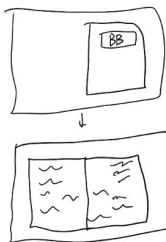
Brainstorming

Phone

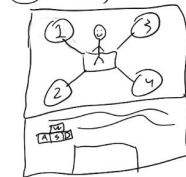


PC/Tablet

① Virtual Textbook



② Learning Game

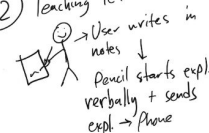


Physical Device

① BrainBook Foldable

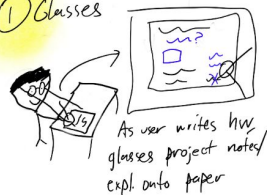


② Teaching Pencil

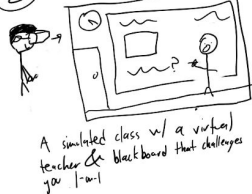


AR/VR

① Glasses



② Virtual Class



Smart Watch

① Quick Verbal Assistant



Device Sketch

Concept Sketches



User casually discovering
on their couch



User using it to
help process a lecture



User uses BB to
synthesize their textbook
& HW



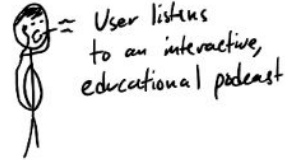
User uses it to
learn a school subject
independently

Voice Memo Sketch

Voice Memo Concept Sketches



User converses w/ it
while pacing



User listens
to an interactive,
educational podcast

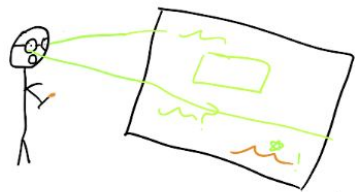


User speaks w/ it less actively
while doing something

AR/VR Glasses Sketch

AR Glasses

Concept Sketch



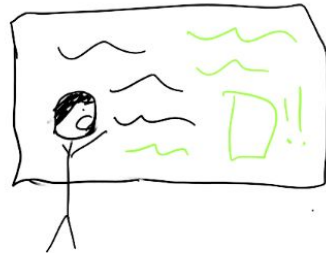
User "projects" onto a blank space to learn in a free space w/ large graphics



User sees interactive annotations & notes on their notes or a textbook



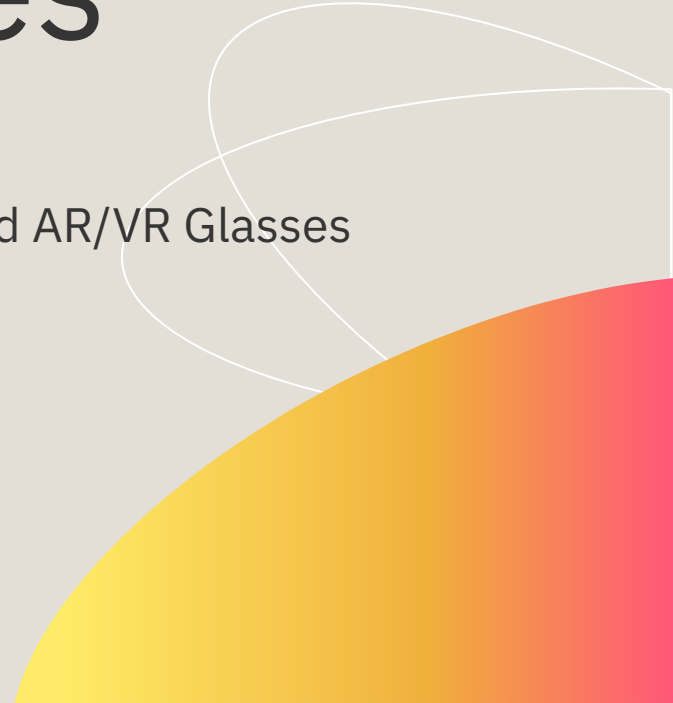
The glasses project quiz problems onto a blank sheet of paper for the user to answer.



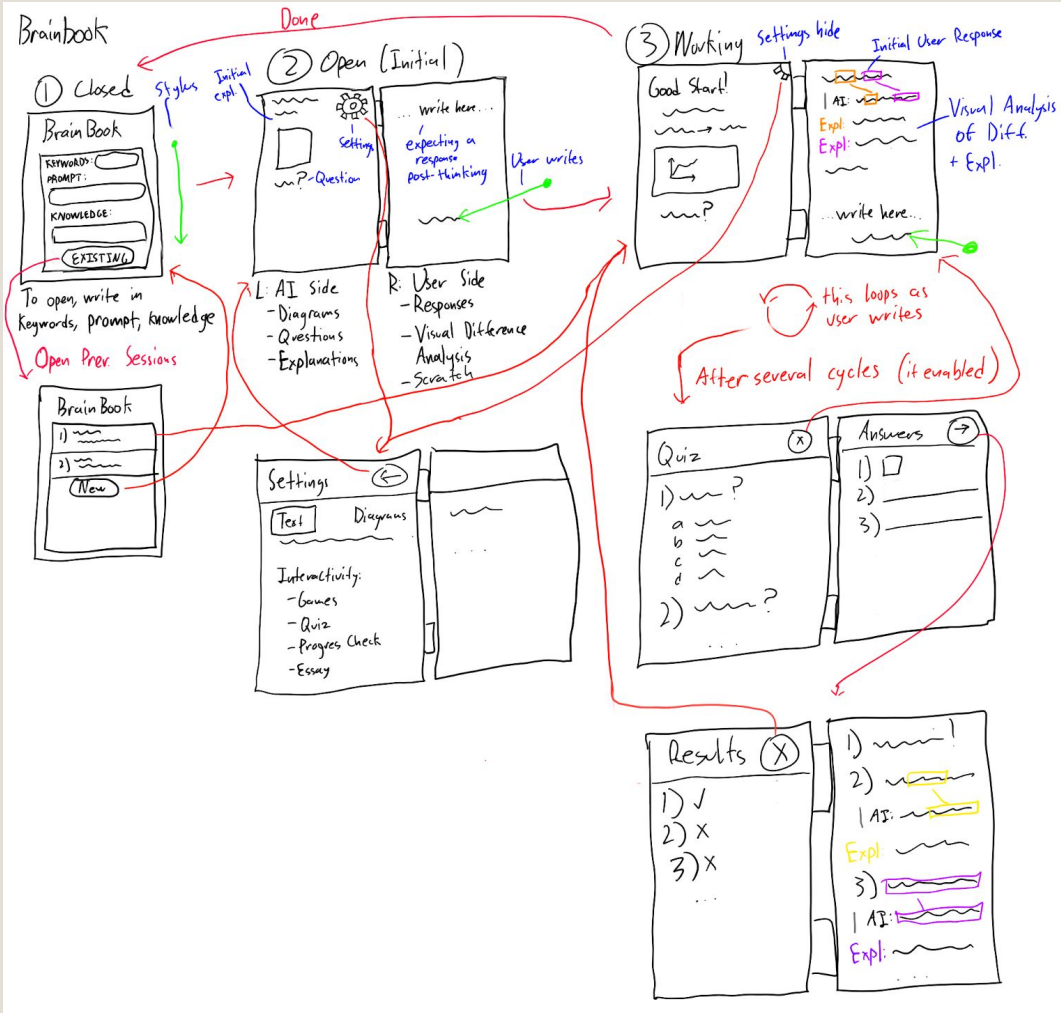
The glasses project extra info or notes during class based on what helps the user the most

Key Frames

Fleshing out sketch ideas of the Device and AR/VR Glasses



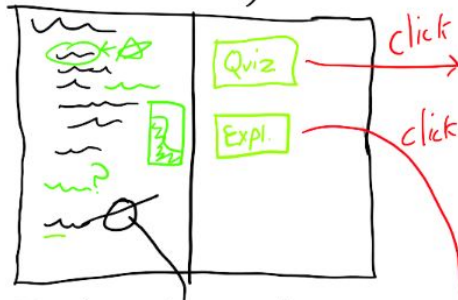
Device



AR/VR Glasses

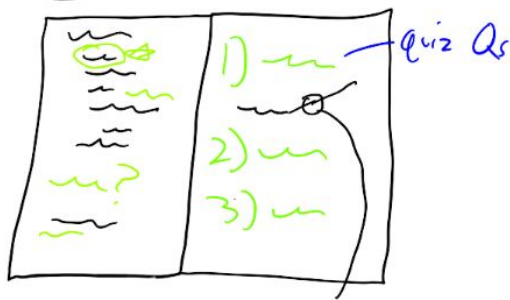
Frames

① Note-writing

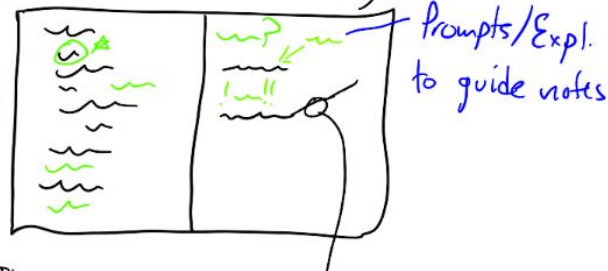


Tap the Glasses to Activate

② Quiz Mode



③ Guided Note-taking



The user continues to write w/
pointers from the AI!

Pros and Cons of the UIs

Device

Pros:

- Intuitive physical design encourages student-guided learning and is familiar
- Spacious, large screens are more accessible and allow more content to be shown

Cons:

- Requires purchase of an external device that the user needs to carry around
- Some may prefer writing with paper or typing
- Needs to be charged

AR/VR Glasses

Pros:

- Extremely portable; works on whatever medium the user is working on
- Can play guiding videos or give explanations while the user is working, which can be helpful

Cons:

- Requires purchase of expensive AR glasses
- AR technology may glitch while the user is writing over overlays

Rationale behind choosing the device

The physical device's spacious and intuitive design would encourage learning and give the user a concrete platform to learn on. By contrast, the AR glasses, which as a technology is still very immature and glitchy, is purely an overlay and so can only ever be a digital assistant rather than a physical part of the user's learning experience. The device is more stable and a better option overall, which is why we chose it.

Prototype Construction

Construction

To emulate a paper textbook, the prototype was made out of paper

- Two papers were taped together to make an openable book

Each unique screen was made on separate pieces of paper

- This makes it easy to switch screens during the experiment by sliding the appropriate paper in
- The contents of each screen were drawn with a pen

Operation

Each interview was conducted by one team member each

- Each team member operated as both the AI and the notetaker
- This is feasible because notes could be written while the user writes

Each interview was conducted in person with a copy of our paper prototype

Ultimately, both the user and the AI wrote directly onto either the book or a specific screen (i.e. the front page, the user/AI space, etc.)

Prototype Construction

Features

- A front page with three starting prompts
 - This must be filled out for the textbook to give a response when opened
- A Settings Screen
 - Accessed via a Settings Icon
 - Closed via an X Icon
- An AI “space”, where the AI writes
- A User “space”, where the user can write, take scratch notes, and see the visual differences of the AI’s & user’s responses

Interactions

- The user writes directly onto:
 - The front page of the textbook
 - The User Space Screen
 - The Settings Screen
- The AI writes directly onto:
 - The AI Space Screen
 - The User Space Screen
 - Solely for Visual Difference Analysis
- The user can press:
 - The Settings icon
 - The X icon on the Settings page
- The user can physically swing the book open and closed

Prototype

Brain Book

KEYWORDS :

QUESTION:

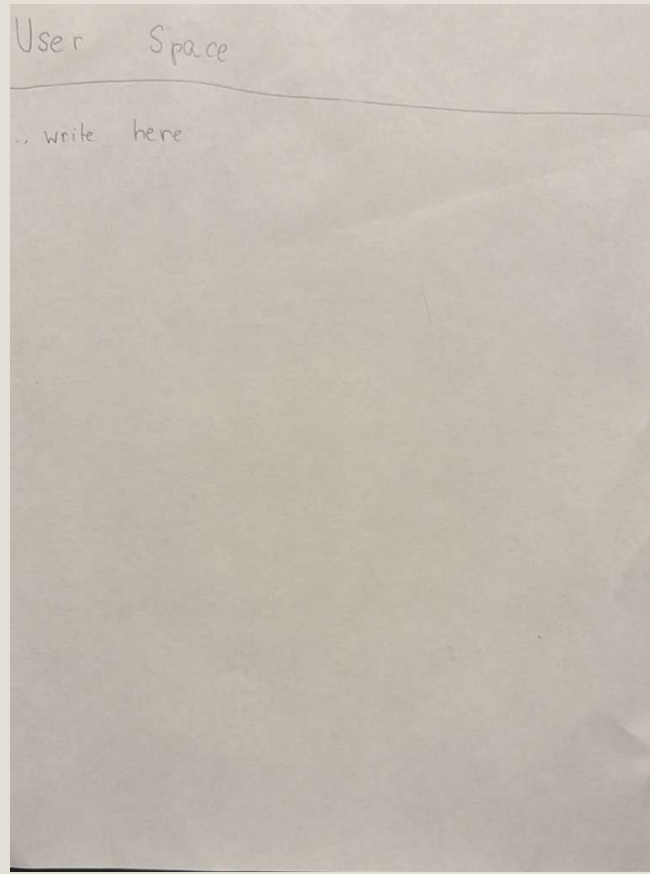
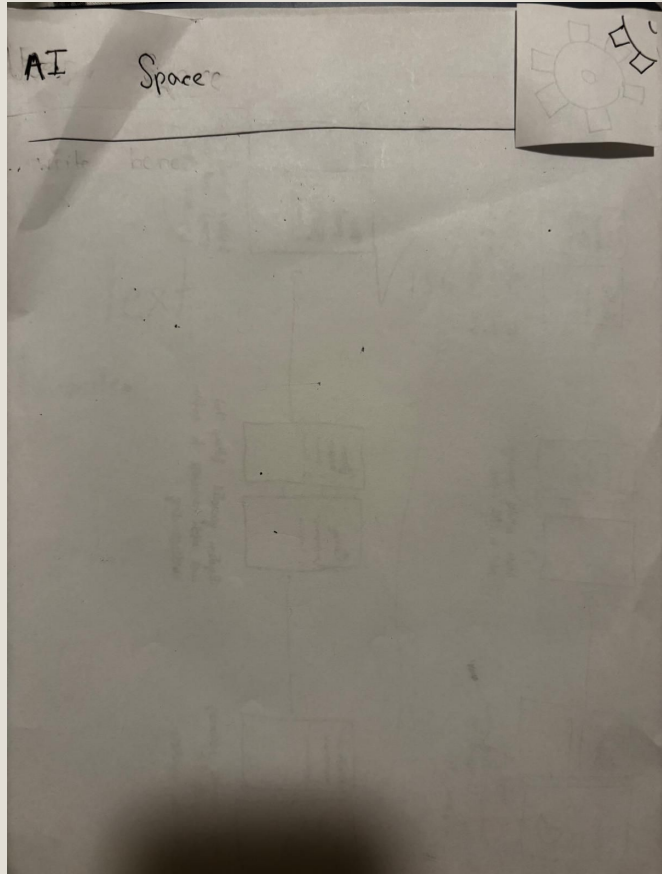
Knowledge:

Existing

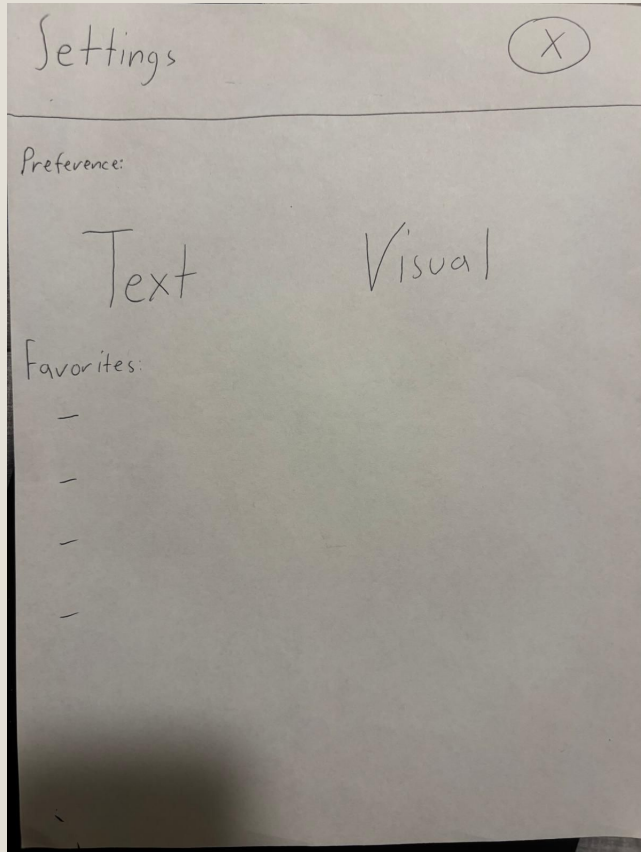
No topic selected
CLOSE ME!

Please describe a new subject or
select an existing one, on the front
page.

Prototype

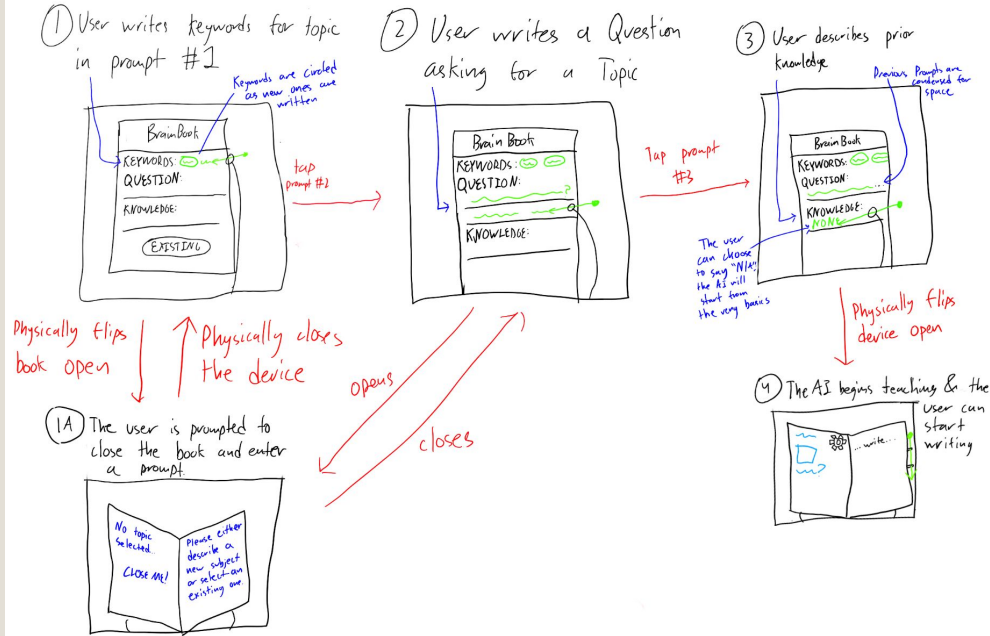


Prototype



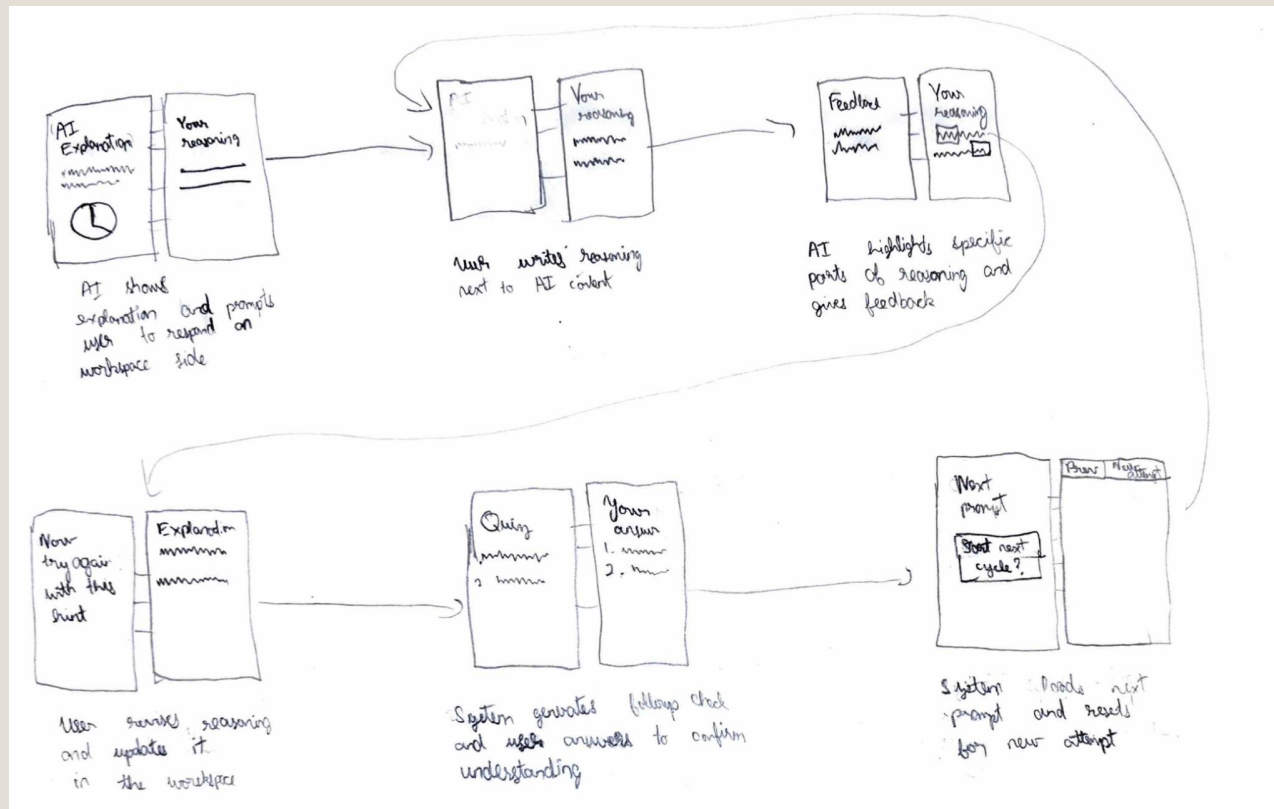
Task Flows

Task #1 Flow



Task #1 Flow (Simple) - The User can fill out the starting prompts to open the textbook

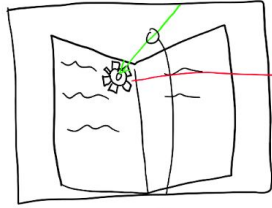
- Opening the textbook before opening the textbook will prompt the user to close the textbook and fill out the starting prompts



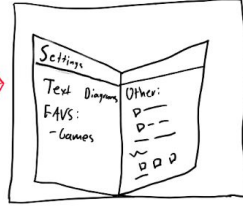
Task #2 Flow (Moderate) - The user can update the user space with a response to the AI's prompting and engage in a learning loop with the AI

Task #3

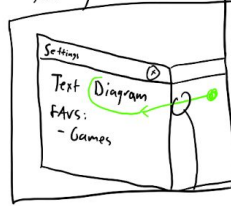
① User taps setting icon



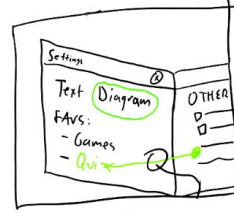
② Textbook shows all settings as-is



② User edits a setting lie, circling diagram to prefer it)

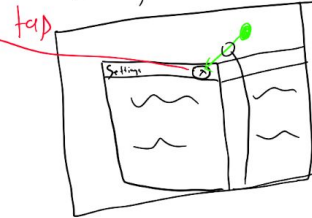


2A) User edits another setting



③ User edits an arbitrary number of settings

④ User clicks (X) to exit settings menu & continue learning



Task #3 Flow (Complex) - The user can change their settings by accessing the settings page, updating the desired settings, and returning to the Main Page

Testing Methodology

Participants

Our primary demographic are **students**, so we tested with students at UCSB and UCLA

- Recruited via Connections
- No Compensation

Environment & Apparatus

We conducted our interviews *in-person*

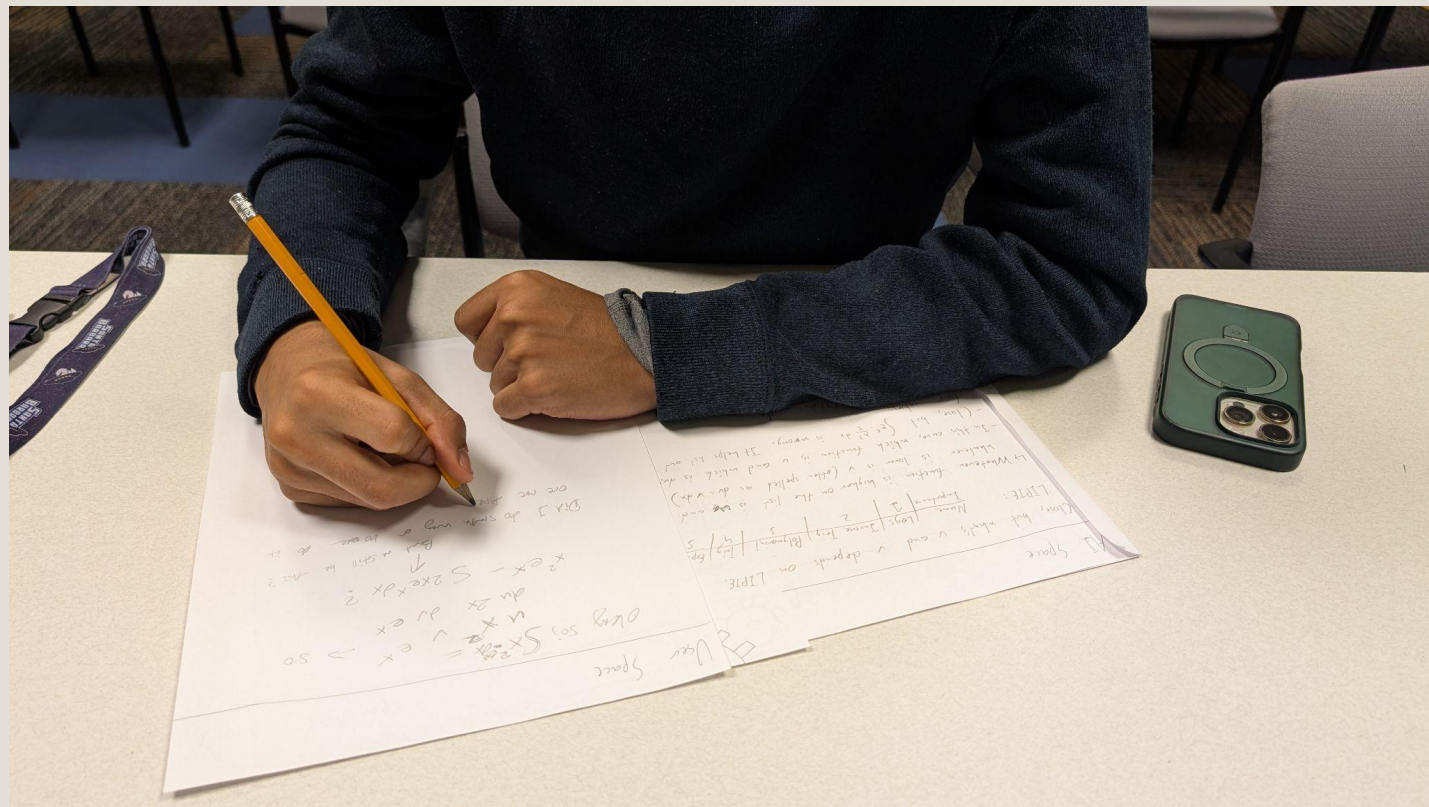
We used a physical paper prototype that the AI & User interacted with using pencils

Procedure

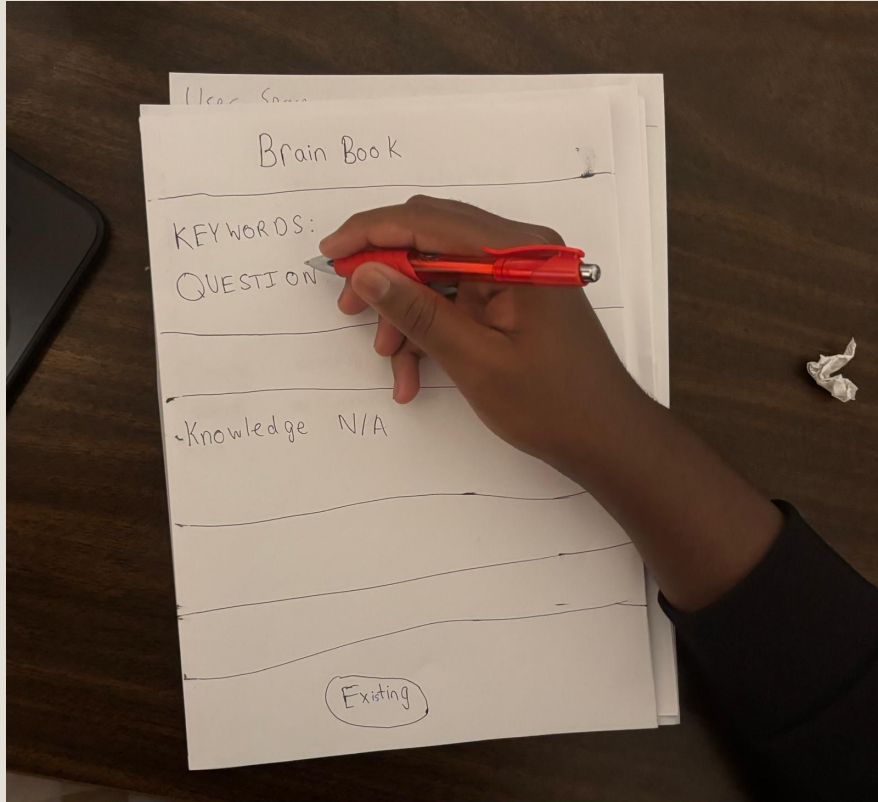
Each team member operated as both the AI & a notetaker

- Interviews were conducted by one team member
- This was feasible because we could take notes while the user wrote

User Interactions with the Prototype



User Interactions with the Prototype

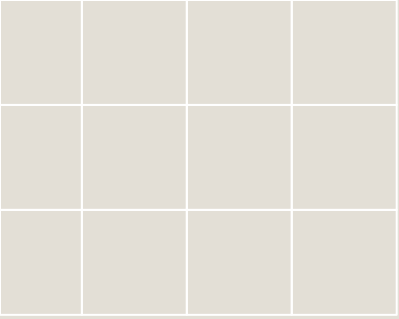


Procedure Description

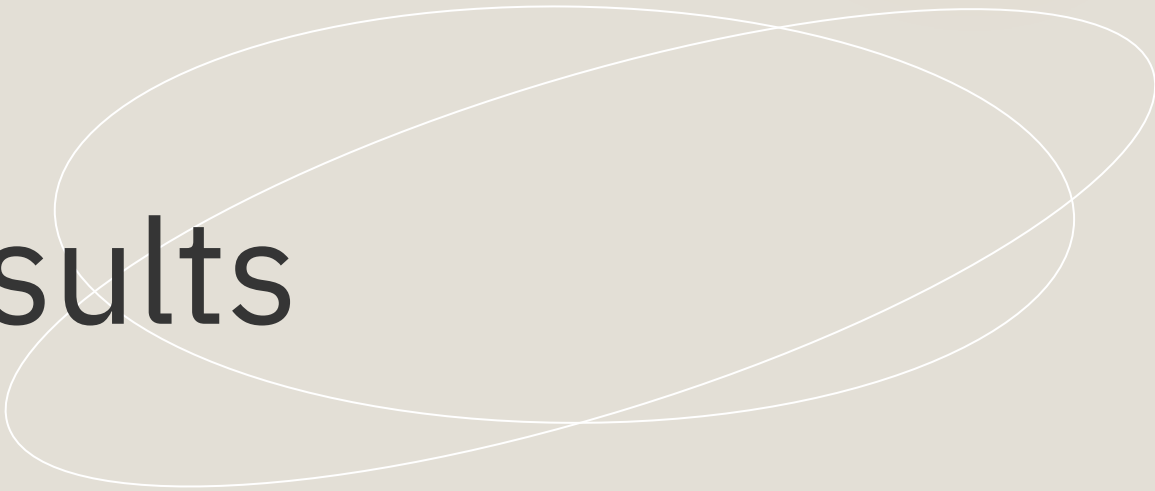
First, show the user where they can enter text and what buttons they can tap. Then, instruct them to query the textbook about “Calculus” (If the user tries to open the textbook without filling out the three starting prompts, show the “User Input Required” Screen). Then, once the user opens the textbook, the AI should quickly write out a basic explanation and a question to ask the user. Then, ask the user to start the learning loop by responding to the AI (if the user tries to write in the AI space, tell them that they can’t write there). The AI should then quickly respond to the user, beginning the learning loop of the user and AI exchanging questions and answers. At some point in the middle of the learning loop, ask the user to change their settings to prefer Diagrams – future responses should then have more diagrams. Eventually, the user should conclude that they understand the topic. Once this occurs, collect final user feedback and answer any questions.

Usability Goals

1. The user should intuitively know how to interact with the textbook without much help
 - a. **Key Measurement:** number of times the user misclicks or attempts to do something that they aren't supposed to do
 - b. **Other Measurements:** The user can smoothly operate the textbook and the various screens without much hesitation
2. Reduction of errors as the product is used
 - a. **Key Measurement:** number of times the user makes a mistake their response
 - i. This should decrease as time goes on
 - b. **Other Measurements:** The user feels like they understand the topic more as they use the product



Testing Results



Process Data

Experiment #1

Overall, the user intuitively understood most of the UI.

They quickly figured out the core learning loop of the product, and were able to learn about Calculus 1 using the product.

While the user did make mistakes when responding to the AI, their understanding solidified until they understood it.

The user wasn't sure how to stop using the textbook initially, and they thought that some of the explanations were unclear or bad.

Experiment #2

User generally understood the UI.

They completed the starting prompts before opening the book, but the prompts' purpose was initially unclear.

They quickly grasped the core learning loop, consistently writing in the correct space.

User was satisfied with the AI feedback once it provided clearer guidance and diagrams.

The user wasn't sure how to stop using the textbook initially

Bottom-line Data

Experiment #1

The user made a single usability error at the end

- They didn't know to close the book and tried clicking the settings dial
- Beyond that, the user did not make any other usability errors

The user made 4 mathematical errors in their responses

- These errors were spread out throughout the experiment, so the rate of errors didn't really decrease
- The user did get the right answer multiple times despite these errors

Experiment #2

The user had one major usability issue at the end.

- They didn't know they were supposed to close the book to end the session and looked for an in-app way to stop.
- They also needed a quick clarification about what the "Favorites" section in Settings meant.

The user made about 3 math errors during the loop, including a wrong choice of u during substitution, but corrected it once they were prompted to explain why it didn't work.

After switching Settings to prefer Diagrams, their responses became more accurate and they completed the later problems without additional mistakes.

Extra Data

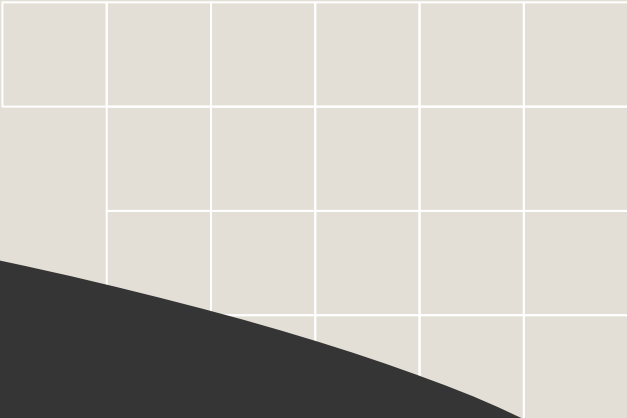
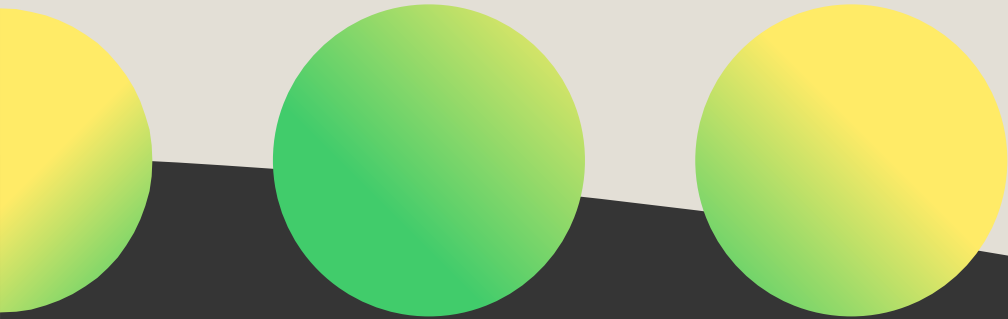
Other Observations

- The front page could probably be a bit more on how to slate the preferred topic
- Ending the product usage could be easier, with maybe an X button for stopping the session
- Overall the AI responses need to just a tad bit more clear, since it can confuse the user and their understanding

Achievement of Goals

1. We definitely achieved goal #1 (intuitive user interface)
 - a. Both users almost immediately understood most of the UI
 - b. There were some usability errors in both experiments, but there were far more successful UI interactions
2. We somewhat achieved goal #2 (reduction in response errors)
 - a. Exp.#2 found a reduction in errors
 - b. Exp.#1 did not find a reduction
 - c. Thus, we somewhat achieved this goal

Discussion



Implications

In both experiments, participants understood how to start, stayed in the correct writing space, and engaged the learning loop with minimal help.

When diagrams enabled, later responses became more accurate and explanations in Experiment #2.

Both participants looked for an on-screen way to stop, instead of understanding that closing the book ends the session.

One participant explicitly flagged unclear AI explanations and asked for clearer intermediate steps between their work and the AI response.

Future Changes

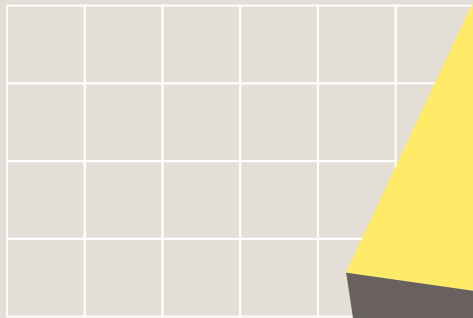
Based on the users' feedbacks & usability errors, we will...

- Add a physical button on the side of the device, similar to an iPad, to turn off the device intuitively
- Change the wording of the Settings Page to say 'Favorite Tools' instead of 'Favorites'
- When the user first activates the device, they will be shown a quick onboarding guide that shows them how to start asking about a topic
 - This onboarding guide will cover starting a new topic, accessing old ones, changing various settings, and interacting with the AI
 - The product will also have a user manual accessible from the settings page

What Testing Couldn't Reveal...

- 1) The interface/testing setup is limiting and can't fully capture how deeply the user actually understood the topic (it mostly reveals whether they can follow the flow and produce an answer).
- 2) With only a small number of participants, we can't confidently generalize the results to different learning styles, skill levels, or broader student populations.

Appendix



All Pros & Cons

PC/Tablet

Pros:

- Intuitive physical design encourages student-guided learning and is familiar
- The stylus requires writing, which is better for learning
- A textbook is familiar to students
- Separate AI / User sides encourages user thinking with AI support
- The user has space to think and externalize while reading what the AI wrote
- Spacious, large screens are more accessible and allow more content to be shown
- Increases what users can read and learn at once
- Diagrams, quizzes, readings, and more can all be shown at once, giving the user a plethora of options to learn
- Requiring three prompts to “open” the textbook gathers a rich amount of context from the start, which will allow the AI to give the best responses

Cons:

- Requires purchase of an external device that the user needs to carry around
- Stylus may get lost, and the foldable is not as durable as traditional electronics
- Not all users may like writing with a stylus
- Some may prefer writing with paper or typing
- Needs to be charged

AR/VR Glasses

Pros:

- Extremely portable; works on whatever medium the user is working on
- Can project onto notebooks, lectures, textbooks, etc.
- Doesn't even need a medium! Can work when the user isn't writing
- Integrates easily and instantly into a user's existing workflow
- No product setup, extra notebook, etc.
- Once the user wears them, they can study as normal
- Annotations show up on existing note taking materials
- Expands upon student learning
- Truly student-driven
- Annotations directly on the notes are more precise and more helpful
- Can play guiding videos or give explanations while the user is working, which can be helpful

Cons:

- Requires purchase of expensive AR glasses
- Can become intrusive or unhelpful
- “Invades” the user's notes, which can be messy
- AR technology may glitch while the user is writing over overlays
- If existing notes are not readable or unreliable, then the AI will be unable to collect enough information

Procedure Script

Script: After demoing where the user can enter text and click buttons:

1. Ask the user to query about “Calculus I” and open the textbook
 - a. If the user tries to open the textbook without entering all three fields, show the “User Input Required to Start” screen
 - b. Let the user write in the three starting prompts
2. Once the textbook is open, the AI should quickly write out a basic explanation and a question to ask the user
 - a. The user will have to wait while this is done
3. Ask the user to begin interacting with the textbook and begin learning
4. The user should write down their response in the User Space
 - a. Don’t tell them where to write
 - b. If the user tries to write in the AI space, tell them they cannot write there
5. The AI should quickly write another response to the user’s response and ask another question
6. The user should understand at this point the general feedback loop of the app, and respond to the AI
7. Repeat steps 5 & 6 until the user says they understand
 - a. To gauge this, ask the user after each iteration if they understand
8. In between iterations of the loop, ask the user to change their settings and select Diagrams
 - a. Don’t tell the user how to do this
 - b. If they misclick, gently guide them to the right path
9. Once the user says they understand the topic, end the survey
10. Collect final user feedback & answer any questions

Critical Incidents #1

Task	Description	Type	Severity	Explanation
#1	User intuitively understood how to start the app (by filling out the 3 starting prompts and then opening the book)	Positive	0	The user immediately filled out the three prompts and opened the book.
#2	User did not attempt to write in the AI space and only wrote where they were able to	Positive	2	The user immediately wrote in the User Space
#2	User engaged the learning loop quickly and engaged completely	Positive	0	The user spent the majority of the experiment reading & responding to the AI

Critical Incidents #1

Task	Description	Type	Severity	Explanation
#3	User fluently navigated to & from the settings page and changed settings easily	Positive	0	The user knew intuitively how to access the Settings page, knew to circle the Diagrams, knew to add various tools, and knew to return from the Settings page
#3	User required clarification on what the "Favorites" section of the Settings page meant	Negative	2	"What's favorites?"

Critical Incidents #1

Task	Description	Type	Severity	Explanation
#2	A few AI responses were unclear to the user, requiring further prompting	Negative	3	The user wrote down "Your explanation is unclear..." and asked for clarification - They got more confused
Ending the Product	The user didn't know how to end their use of the product	Negative	3	The user needed an explanation to realize they needed to close the book - They tried clicking the Settings dial like it was an X button

Critical Incidents #2

Task	Description	Type	Severity	Explanation
#1	User intuitively understood how to start the app (by filling out the 3 starting prompts and then opening the book)	Positive	0	The user filled out Keywords, Question, and Prior Knowledge first, then opened the book without attempting to bypass the start flow.
#1	User was unsure what the 3 starting prompts meant at first	Negative	2	The user didn't misclick, but needed a moment to interpret what each prompt was asking before entering the "right" kind of information.
#2	User did not attempt to write in the AI space and only wrote where they were able to	Positive	0	The user immediately wrote in the User Space

Critical Incidents #2

#2	User made early response mistakes and needed targeted feedback to correct reasoning	Negative	2	The user made an error on the first question, then made another mistake because they didn't connect prior knowledge to a key step until the AI showed how to link it.
#2	User engaged the learning loop quickly and engaged completely	Positive	0	The user spent a majority of the experiment reading & responding to the AI
#3	User fluently navigated to & from the settings page and changed settings easily	Positive	0	The user knew intuitively how to access the Settings page, knew to circle the Diagrams, knew to add various tools, and knew to return from the Settings page

Critical Incidents #2

#3	User required clarification on what the “Favorites” section of the Settings page meant	Negative	2	“What’s favorites?”
#2	AI responses were clear to the user	Positive	0	The user was satisfied with the ai explanation and feedback, and understood clearly
Ending the Product	The user didn’t know how to end their use of the product	Negative	3	The user needed an explanation in order to realize they needed to close the book



Thank you